

Param Mankad

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EDUCATION

Nirma University, Institute of Technology

Bachelor of Technology – Mechanical Engineering

Ahmedabad, Gujarat, India

2023 – 2027

- **CGPA:** 9.42 / 10.0 **Batch Gold Medalist**
- **Minor:** Data Science
- **Relevant Coursework:** Computational Fluid Dynamics, Fluid Mechanics & Turbulence, Numerical Methods, Heat Transfer, Finite Element Analysis, Machine Design, Dynamics of Machines, Basics of Flight & Aerodynamics

RESEARCH EXPERIENCE

Multiphase Electrohydrodynamics – CFD Research, OpenFOAM

2025 – Present

Nirma University, under Prof. Absar Lakdawala

- Extending the *interEHDFoam* solver (EHD variant of *interFoam*) to couple the energy transport equation with multiphase electrohydrodynamic flow, enabling thermally-coupled EHD simulations.
- Implemented finite-volume discretization of the sensible enthalpy transport equation within the *interEHDFoam* solver architecture, including diffusive and convective flux assembly and boundary condition handling.
- Investigating EHD body-force-driven interfacial instabilities, bubble morphology deformation under external electric fields, and velocity-field coupling in confined capillary geometries (Taylor flow regime).
- Analysing numerical convergence behaviour, time-step stability ($Co \leq 0.5$), and iterative solver performance for coupled pressure-velocity-electric field solutions.

RESEARCH PUBLICATIONS

Conference Paper

- *Design Evolution and Aerodynamic Performance Assessment of a Micro-Class Fixed-Wing Unmanned Aerial Vehicle (UAV)*
Presented at the **12th International & 52nd National Conference on Fluid Mechanics and Fluid Power (FMFP 2025)**, Nirma University. Published in conference proceedings.

Journal Paper (Under Preparation for Submission)

- *Aerodynamic Performance Assessment of High-Payload Fixed-Wing UAV*
Manuscript in preparation for submission to the *Journal of Flow Visualization and Image Processing*.

EXPERIENCE

SAE Nirma Collegiate Club – Team Arrow

Nirma University

Analysis & Technical Lead

Mar 2025 – Feb 2026

- Led computational aerodynamic analysis for multi-rotor drones and fixed-wing UAV platforms using ANSYS Fluent and XFLR5.
- Directed CFD-informed structural feasibility assessments, weight budget allocation, and aerodynamic efficiency trade-offs across iterative design cycles.
- Produced and interpreted pressure-coefficient distributions, lift/drag polars, and wake-interaction studies to guide geometry refinement.

Vice Captain

Oct 2024 – Feb 2026

- Coordinated UAV development for SAE Aero Design 2026 (Advanced Class) and SAEISS Drone Development Challenge 2025, overseeing design reviews, system integration, and competition preparation.

Analysis Engineer

Oct 2024 – Mar 2025

- Conducted aerodynamic analysis and performance evaluation of UAV configurations; supported design decisions through CFD results and engineering calculations.

Mentor

Feb 2026 – Present

- Mentoring undergraduate teams in UAV aerodynamics, CFD workflows, and competition strategy for national and international

events.

Development Team Member

Aug 2023 – Oct 2024

- Contributed to CAD modelling, airframe fabrication, system integration, and ground testing of competition UAVs.

SIGNIFICANT ACADEMIC PROJECTS

Development of Multiphase Electrohydrodynamics Solver with Energy Equation in OpenFOAM

Ongoing

- Extending the in-house developed *interEHDFoam* solver to incorporate the energy transport equation.
- Implementing finite volume discretization of the sensible heat equation for coupled electrohydrodynamic–thermal multiphase simulations.
- Enabling simulation of temperature-dependent electrohydrodynamic flow phenomena.

Numerical Analysis of Taylor Flow in Capillary under Electric Field

Ongoing

- Simulating Taylor bubble flow dynamics in capillary channels subject to external electric fields in OpenFOAM; resolving interfacial deformation, bubble elongation, and velocity-field redistribution under EHD forcing.
- Studying coupling between Maxwell stress-induced body forces and multiphase flow topology in confined, low-Gravitational Bond-number geometries.

Analysis & Autonomous System Development – Hybrid VTOL UAV

May 2025 – Feb 2026

- Performed aerodynamic and structural validation using full-configuration CFD (ANSYS Fluent, rotating propeller MRF sub-domain at ≈ 26000 RPM) and FEA.
- Analysed propeller-wake/lifting-surface interaction during VTOL-to-fixed-wing transition; quantified induced drag and pitching-moment perturbations.
- Developed autonomous navigation and mission-execution pipeline using ArduPilot and Mission Planner, including waypoint following and VTOL transition logic.

Design & Development of Micro-Class Aircraft – SAEISS DDC 2025

Oct 2024 – Mar 2025

- Designed and fabricated a micro-class fixed-wing UAV (empty mass 1.15 kg) optimised for maximum payload fraction; achieved payload fraction of 0.72 carrying a 3 kg payload for a 3-minute endurance flight.
- Validated aerodynamic performance via XFLR5 panel-method analysis and ANSYS Fluent RANS simulations; iterated wing planform for C_L/C_D maximisation.

Spur Gearbox Design – Two-Stage 9 : 1 Reduction

- Designed a two-stage spur gearbox (7.5 kW, 1000 RPM, 9 : 1 reduction) using Lewis bending theory with wear-strength (Hertzian contact) verification.
- Optimised module, face-width, and helix-angle selections for safe bending and pitting resistance; completed shaft design under combined bending-torsion loading with key and hub-fitment validation.

TECHNICAL SKILLS

CFD & Numerical Methods	Finite Volume Method, RANS turbulence modelling, Multiphase Flow (VoF), Electrohydrodynamics
Simulation Software	OpenFOAM, ANSYS Fluent, Tecplot, ParaView, XFLR5
Programming & Scripting	Python (NumPy, SciPy), MATLAB, Bash
Engineering & Design	SolidWorks, Fusion 360, ArduPilot / Mission Planner, $\LaTeX$
Machine Learning	Scikit-learn, Linear/Ridge Regression, Naive Bayes (Data Science minor)

HONOURS & AWARDS

- **SUAS 2025** – Global Champions (1st Place)
- **SAEISS Drone Development Challenge 2025** – 1st Place (Regular Class)
- **SAEISS Drone Development Challenge 2025** – 3rd Place (Micro Class)
- **SAEISS Drone Development Challenge 2024** – 2nd Place (Micro Class)
- **SAE Aero Design West 2025** – 5th Place Mission Score